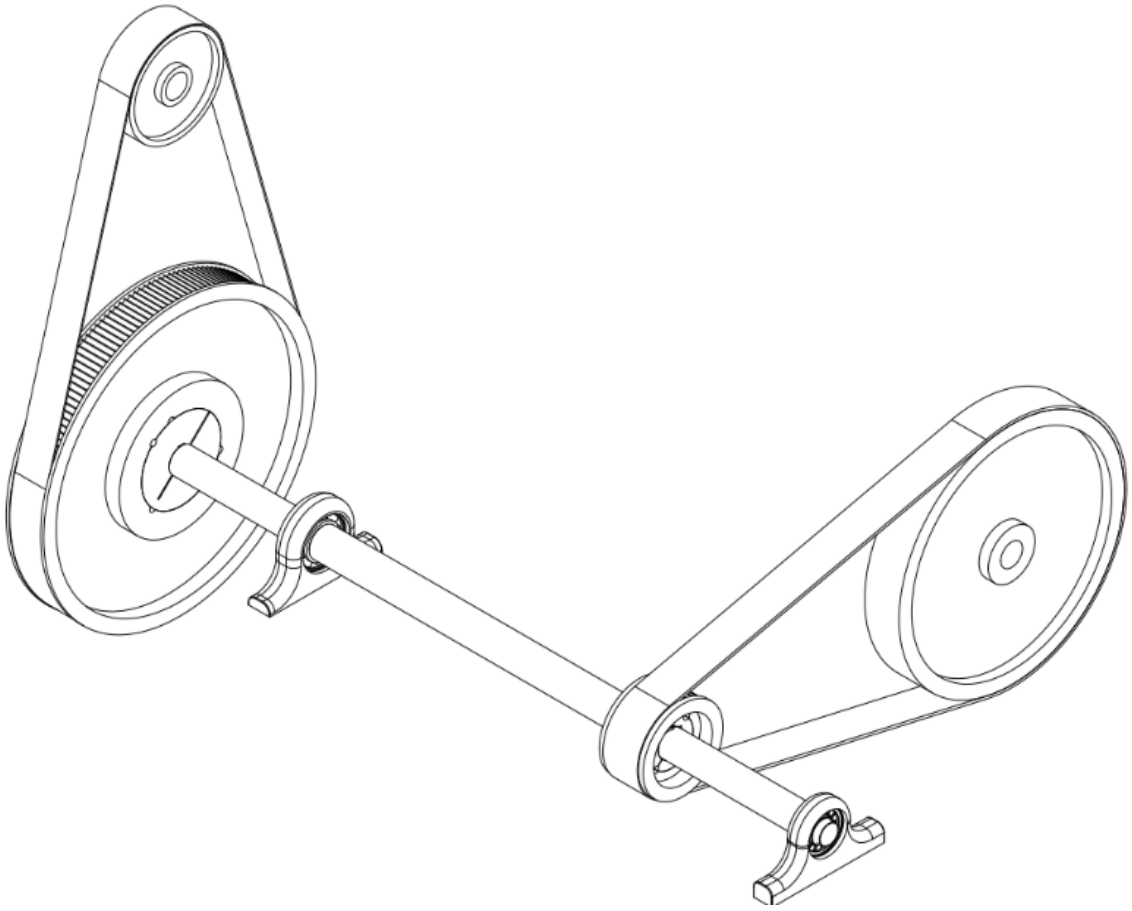
Shaft Design Report

SD – 133

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Summary

The design for a power transmission assembly to power a Positive Displacement multi-cylinder Piston Pump was requested by a client. A layshaft was designed to transmit power from an electric motor (operating at 1440 rpm) to the pump (operating at 150 rpm) as part of a two-stage transmission drive. Reduction ratios and force analysis calculations (shear force, bending moment and shaft loading) were performed to outline the design of the shaft. Shaft layout and individual components were then chosen and iterated upon to produce a design that would not fail under the design loading conditions. Stress analysis was also performed to ensure the smooth operation of the transmission system. A suitable design that meets all requirements and most consumer desires was created, however, a few trade-offs had to be made and recommendations to improve this design are outlined at the end of the report.

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1. Introduction

The design brief consisted of designing a layshaft for a Positive Displacement multi-cylinder Piston Pump that delivers 3 litres per second at 90% efficiency and is driven by an electric motor that spins at 1440 rpm [1]. As this was a one-off project, the shaft can be custom-made to fit the design requirements. An iterative design approach was used to choose the standard components (transmission and bearings) and to determine the size, configuration, and shape of the layshaft. This report includes a full overview of the final design, a discussion of the other design options considered, and recommendations for further design iterations.

2. Final design analysis

The final design is an accumulation of positive attributes of prior iterations which do not include issues identified in each discussion subsection. After all iterations, the final shaft design was created in Inventor. Figure (1) shows 45-degree chamfers on the shoulders and at the end of the shaft to conform with Design Want No.11 [1], whilst also enabling easier assembly. The shaft doesn't require any upfront tooling costs as all dimensions can be achieved by standard tools on a lathe. The shaft is toleranced to an H7, h6 fit, geometrical tolerances were linked to datums set on the bearing positions as they are the shaft's principal reference points.

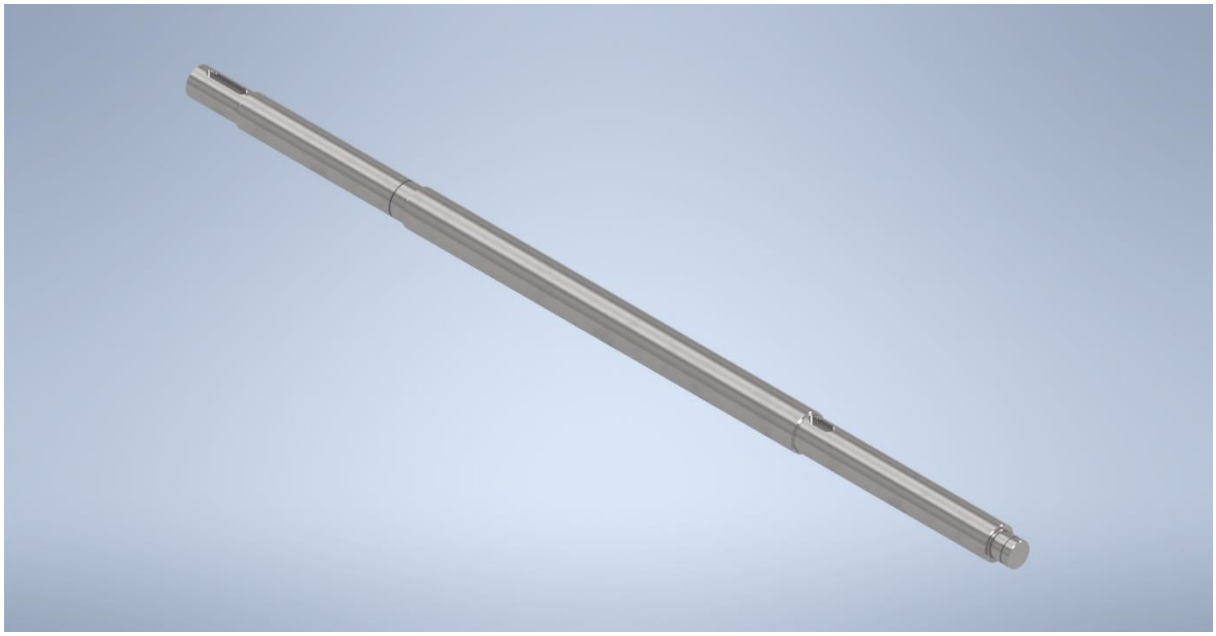


Figure 1: Rendered top view of the final shaft design.

a. Weight, cost and embodied CO_2

The total weight, cost and embodied carbon associated with the final shaft design if shown in Table (1). The material used was Low-alloy manganese-molybdenum steel EN16 605M36 due to its yield strength of 525 Mpa[1].

Table 1: Weight, Cost and CO_2

Weight of shaft (kg)	11.974
Weight of the workpiece (kg) (4mm added to each end)	13.858
Cost (£)	27.02
CO_2 (kg)	27.716

b. Shaft loading conditions and failure testing

The free-body diagrams in Figures (2) and (3) show the loading conditions on the shaft. The results are highlighted in Table (2).

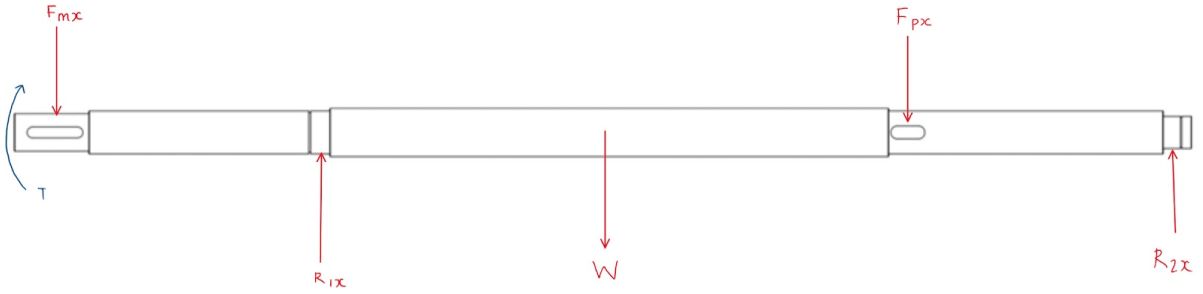


Figure 2: FBD of the final shaft in the x-axis.

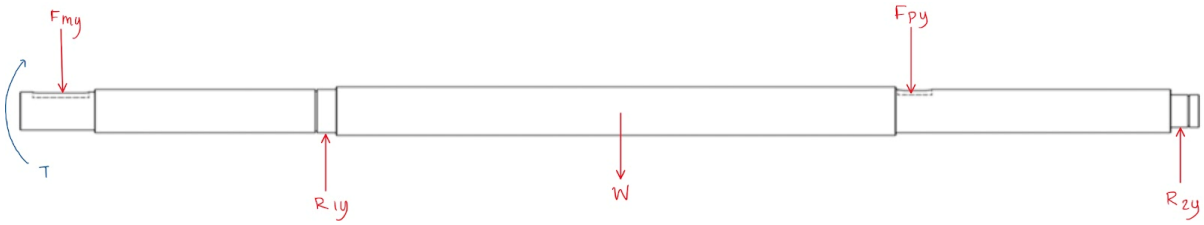


Figure 3: FBD of the final shaft in the y-axis.

Table 2: Results of loading calculations.

F_{mx} (N)	1018
F_{px} (N)	818
R_{1x} (N)	1591
R_{2x} (N)	244
F_{my} (N)	376
F_{py} (N)	3639
R_{1y} (N)	1628
R_{2y} (N)	2377
$F_{m\,total}$ (N)	1085
$F_{p\,total}$ (N)	3720
$R_{1\,total}$ (N)	2277
$R_{2\,total}$ (N)	2390
T_{total} (N)	232

Figures (2), (3) and Table (2) ensure the shaft can sustain these loading requirements and allow BM, SF and Q diagrams to be constructed. The final shaft design will abide by these loading conditions, calculations can be found in Appendix (A).

Creating the bending moment, shear force and torque graphs in MATLAB highlighted potential failure nodes that require testing in the stress calculation spreadsheet so that minimum shaft diameters could be calculated (Figures 4-8).

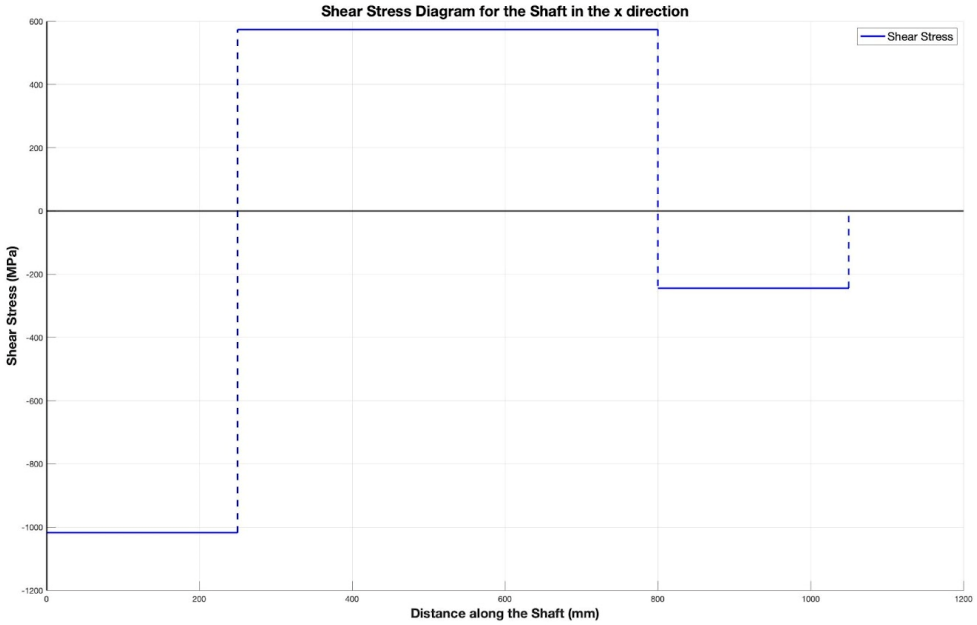


Figure 4: Shaft's Shear Stress diagram in the x direction.

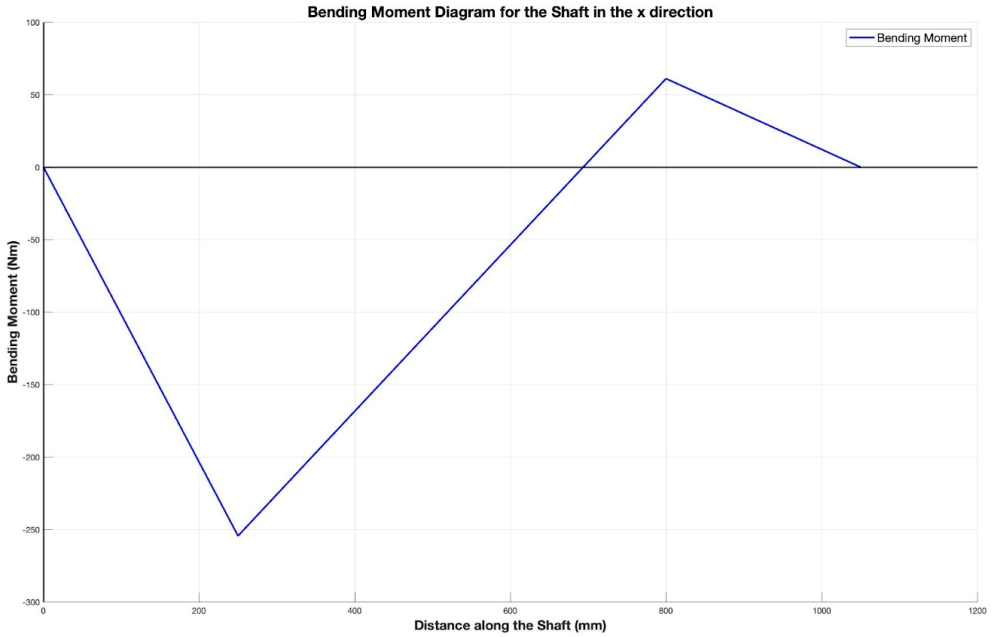


Figure 5: Shaft's Bending Moment diagram in the x direction

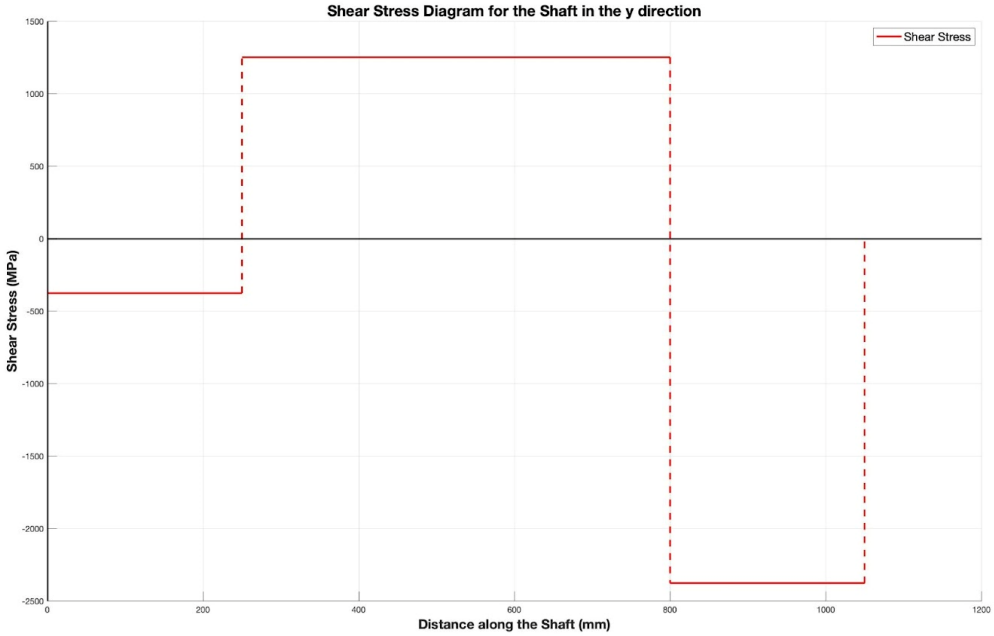


Figure 6: Shaft's Shear Stress diagram in the y direction

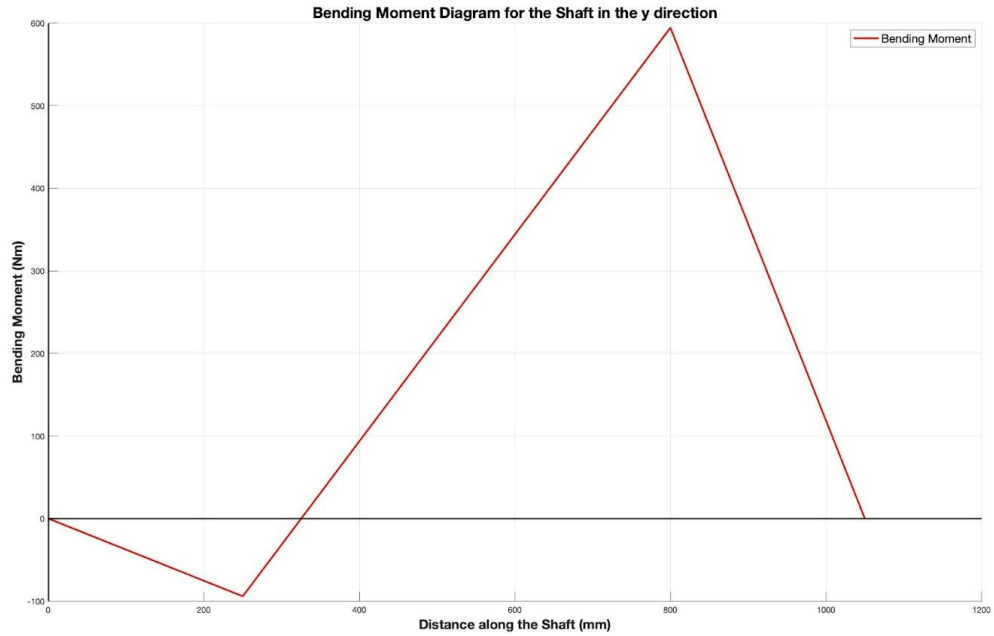


Figure 7: Shaft's Bending Moment diagram in the y direction

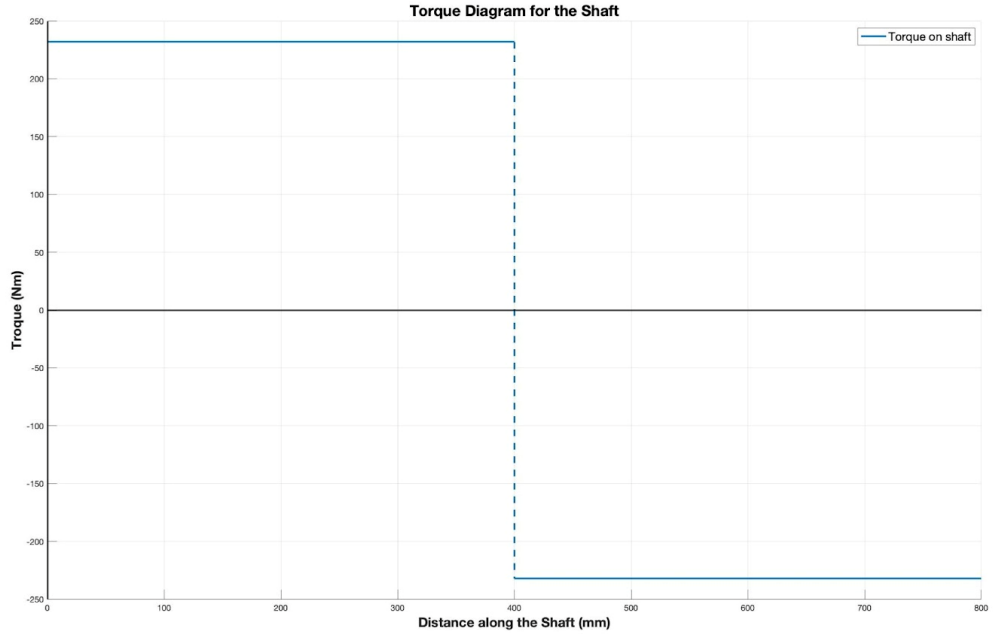


Figure 8: Shafts torque diagram

Figure (4) and Figure (6) show that the shear stress in the x and y direction is maximum within the range 250mm – 800mm, between the two pulleys. This indicates there are potential failure nodes within this axial range.

Figure (5) and Figure (7) show that the bending moment in the x and y directions is maximum at 250mm and 800mm, where the pulleys are located. Indicating that these points carry the largest bending stresses.

Using data from the shear force and bending moment diagrams, along with knowledge of anticipated failure nodes Figure (9) was constructed.



Figure 9: Final shaft iteration with failure nodes indicated.

At these nodes the combined bending moment, torque and appropriate SCFs for the shoulder, circlips and keyways were applied in the failure test equations to identify problematic nodes (Appendix B). SCFs for the key seat and the circlip groove were taken from Shigley's textbook (Table 3) and SCF's for the shoulders were calculated via graphs in Appendix (C).

Table 3: Shingley's bending, torsional and axial SCF's

	Bending	Torsion	Axial
Shoulder fillet – sharp $r/d = 0.02$ ($r/d = 0.02$)	2.7	2.2	3.0
Shoulder fillet – rounded ($r/d = 0.1$)	1.7	1.5	1.9
End-mill keyseat ($r/d = 0.02$)	2.1	3.0	--
Retaining ring groove	5.0	3.0	5.0

Assumptions were made so the nominal tensional stresses in the shaft could be taken as negligible. Therefore the combined stress was constituted only from nominal bending and torsional stresses. A value for the design factor for yield (N_y) was calculated from a combination of a fatigue factor (b), shock factor (c) and Safety factor (d) to give N_y=1.95. A fatigue factor of 1 was chosen as the stress doesn't vary between compression and tension and the shock factor was taken to be 1 as the load is constantly applied. The safety factor was estimated via the Pugsley method, d=XY [2]. The probability of failure (X) was taken to be 1.95 and the seriousness of failure (Y) was taken to be 1.0.

All nodes identified in Figure (9), for the final dimensions of the shaft, were below the 100% maximum allowable stress percentage with the greatest stress percentage of 85.12% at node 5. This final iteration of the stress spreadsheet was formulated to ensure maximum stress percentages were achieved to reduce the weight of the shaft whilst preventing failure (Appendix B).

c. Final component selection

The stress calculation spreadsheet allowed the minimum diameters at various lengths of the shaft to be calculated. From these values pulleys, belts, bearings, taper lock bushes, circlips and shoulder sizes were calculated and chosen from manufacturers' guides to ensure correct sizing to avoid failure (Design Want No.7) [1]. The list of components used in the final design has been tabulated in Table (4).

Table 4: Components used in the final design assembly.

Component	Suppliers No	Description
Bearing A	SKF 6208 2	Deep groove ball bearing
Bearing B	SKF NU 206 ECP 2	Roller bearing
Taper lock for the pump pulley	Fenner taper lock bush metric bore size 2012	
Taper lock for the motor pulley	Fenner taper lock bush metric bore size 3525	
Circlip	Cirteq DIN 471 – 40x1.75	Shaft retaining circlip for Bearing A
Circlip	Cirteq DIN 471 –30x1.5	Shaft retaining circlip for Bearing B
Pump pulley	Fenner Pulley	Fenner HTD timing belt pulley for taper-lock bush section 8M belt width 50 steel with 168 grooves
Motor pulley	Fenner pulley	Fenner HTD timing belt pulley for taper-lock bush section 14M belt width 55 steel with 28 grooves
Pump belt	Fenner Torque Drive Plus 3 14MXP & HTD 14M Drives	Pump belt width 55, length 1890
Motor belt	Fenner Torque Drive Plus 3 8MXP & HTD 8M Drives	Motor belt width 50, length 2000
Pump key	BS 4235 – 1 rectangular	b=8mm h=7mm l=32mm
Motor key	BS 4235 – 1 rectangular	b=14mm h=9mm l=52mm

To ensure the bearing's suitability for the design the following constraints were calculated using equations in (Appendix D) to give parameters shown in Table (5).

Table 5: Parameters for bearing selection.

	Bearing A	Bearing B
L_{10} (millions of revolutions)	630.7	630.7
Dynamic loading C (kN)	19.52	20.50
Static loading C_0 (kN)	2.28	2.39
Minimum internal diameter (mm)	40	30

To ensure the keyways and keys' suitability for the final shaft design Von Mises equations for stress were used to calculate the minimum length of the keys. Equations can be found in Appendix (E). The minimum lengths of keys due to shear stress and crushing from the normal force are shown in Table (6).

Table 6: Parameters for key selection.

	ι_1 (mm)	ι_2 (mm)	<1.5d (mm)	>2b (mm)
Pump key	4.7	8.5	<67.5	>14
Motor key	12.4	16.7	<45	>28

ι_1= minimum length of key required due to shear stress on the key

ι_2= minimum length of key required due to crushing by normal force

To ensure the transmission's suitability for the final design, dimensions for the belt were calculated referring to the Fenner Torque Drive Plus 3 8 MXP, 14 MXP & HTD 8M, 14M Drives catalogue. Table (7) contains important values used to choose transmission pulleys.

Table 7: Parameters for transmission selection.

	Motor to Shaft	Shaft to Pump
Belt pitch	8	14
Speed Ratio	3 : 1	3.2 : 1
Pulley Selection	56 to 168 grooves	28 to 90 grooves
Belt Length / Centre Distance (mm)	2000 / 533	1890 / 513
Power Rating / Belt Width	62.60 kW / 50 mm	21.86 kW / 55 mm
Pulley Catalogue Codes	Motor: 043L0056 Shaft: 043L0168	Shaft: 043P0028 Pump: 043P0090

Each pulley had its respective taper locks which eliminated the need for taper lock selection.

If intended details were not gained from the final design analysis, the discussion section of this report explains all design choices and iterations for all parts of the shaft assembly in more detail.

3. Discussion

a. Choice of arrangement

The design brief contained geometrical constraints for the location of pulleys and bearings on the shaft in relation to one another, outlined in Figure (10), with the only axial location design consideration for the components being the location of the motor pulley.

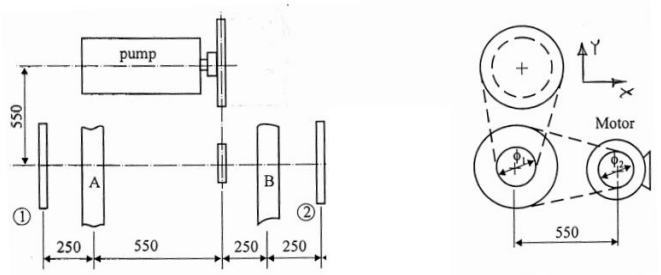


Figure 10: Sketch of the layouts of the pump, shaft, and motor. The motor can be driven at points 1 or 2 and bearings to support the shaft are located at points A and B.

After reviewing initial loading considerations and diagrams of layouts (Figures 11, 12, 13), it was decided to place the motor pulley at location 1 as if placed at location 2 the shaft would be unevenly weighted and prone to higher stresses.



Figure 11: First shaft layout design with motor pulley located at point 1.

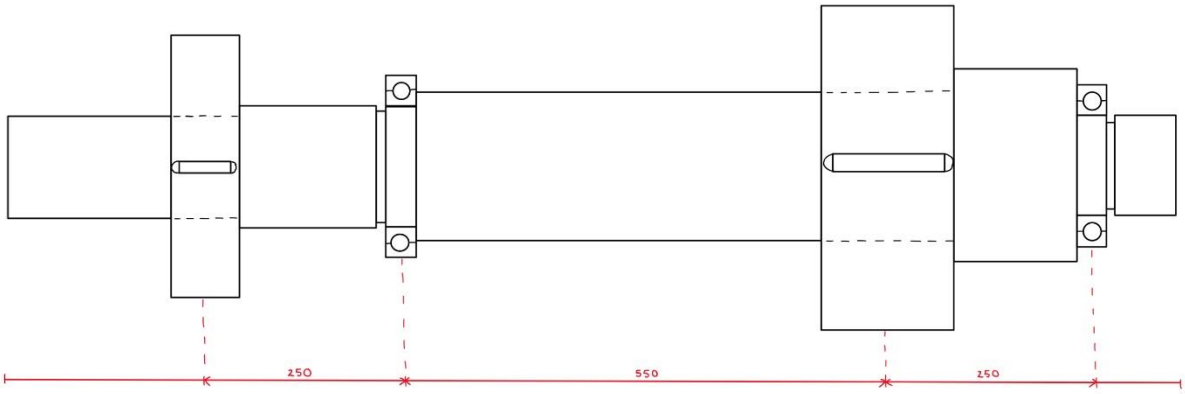


Figure 12: Second shaft design layout with motor pulley located at 1.

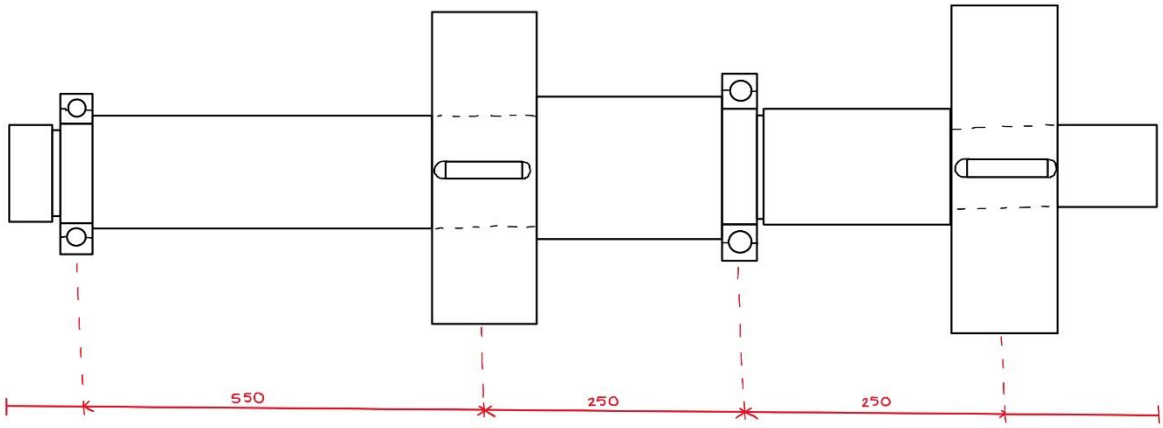


Figure 13: Shaft design layout with motor pulley located at 2.

Due to the increased stress in the shaft in Figure (13), the yield strength of the shaft material would have to increase accordingly, making it more expensive and increasing the approximate embodied CO2, as seen in Table (8). Therefore, the only shaft designs iterated further were those in Figures (11) and (12). The shaft in Figure (12) would be the more expensive of the two, this is due to the initial size of the workpiece needing to have a diameter of 50mm rather than 45mm. The increase in the size of the diameter would also increase the approximate embodied carbon associated with the shaft.

Table 8: Available steel grades in the design briefing document

BS970:1955	BS970:1991	Type of steel	Yield strength (MPa)	Cost (£/kg)	Approx embodied CO2 (kg/kg)
EN3A	070M20	General purpose mild steel	215	1.25	1.7
EN8	080M40	Carbon manganese steel	280	1.36	1.8
EN16	605M36	Low alloy manganese-molybdenum steel	525	1.95	2.0
EN24	817M40	Medium carbon low alloy steel	680	2.66	2.1

Both shaft geometries, in Figures (11) and (12), allow all components to be assembled and disassembled from the ends of the shaft (Design Demand No.18) [1]. However, the shaft in Figure (11) was more favourable to assembly and disassembly of transmission elements from the shaft that may require maintenance, without full disassembly (Design Want No.24) [1]. Whereas the shaft in Figure (12) must be assembled via one end of the shaft, making it difficult to replace components located further in the shaft. This reason paired with the fillet issue in the spreadsheet resulted in the discontinuation of this shaft layout. The largest shaft diameter in Figure (11) doesn't contain any components and the diameters

decrease from the central point meaning the only sliding over a tight fit of significance is for Bearing A. This distance was difficult to reduce due to the motor pulley being axially located via a shoulder and a taper. To reduce the length of sliding over tight fits a circlip on either side of the motor pulley could reduce the length that bearing A would move over if the shoulder was added directly before the component to accommodate the width of the bearing. This however would decrease the manufacturability of the shaft and increase unnecessary details. Another solution for this design problem would be to tolerance the shaft to be in a smaller diameter range before the bearing circlip and have normal tolerancing after the circlip.

b. Transmission

Several factors had to be considered when selecting a suitable power transmission option: power and torque transferred, motor and pulley operational speeds, and geometrical limitations. The motor rotates at 1440 rmp and the pump has an operating speed of 150 rmp, so a total reduction ratio of 9.6:1 was needed to ensure Design Demand No.2s success [1]. Severa reduction ratio combinations were considered. The first ratios considered were 5:1 (motor to shaft) and 2:1 (shaft to pump) but this arrangement was unsuitable due to unbalanced forces, leading to bearing A being under a much greater load than bearing B. So, a more balanced combination was chosen: 3:1 from the motor to the shaft and 3.2:1 from the shaft to the pump.

Belts and pulleys were considered viable transmission options both with their respective advantages and disadvantages. Initial calculations were made for both transmission options leading to belts being selected; a chain drive was not a viable option at this power level. A Fenner 12B could not take the load without being in a triplex configuration whilst the 16B chain was excessive for the requirements of the layshaft. This would have resulted in an expensive transmission option as chain drives have higher installation, maintenance, and production costs, as well as being less environmentally friendly than a belt drive. Belts also better conform to the wants of the customer as they are quieter than chain drives and have a higher durability rate. A limitation of belts is their operating temperature, but the required working temperature (-15°C to 50°C from Design Demand No.8) is within Fenner's *Torque Drive Plus 3 (TDP3)* belt operating temperatures.

Fenner's TDP3 belts, or designated MXP, are Fenner's next-generation belt line. They are compatible with standard pulleys making them highly versatile and compatible with standard systems and are available in 8 and 14-mm pitches. These belts allow for lighter, more compact, and cost-effective drive systems to be designed as well as having a longer service life. The service factor calculated was 1.5 which indicates a soft start medium-duty belt operation, this was used for both sets of belts. The pump requires 11,666.6 W to operate leading to a total design power of 17,499 W. The motor-to-shaft belt was the first to be selected. After looking at Figure (14) the 8MXP belt was chosen.

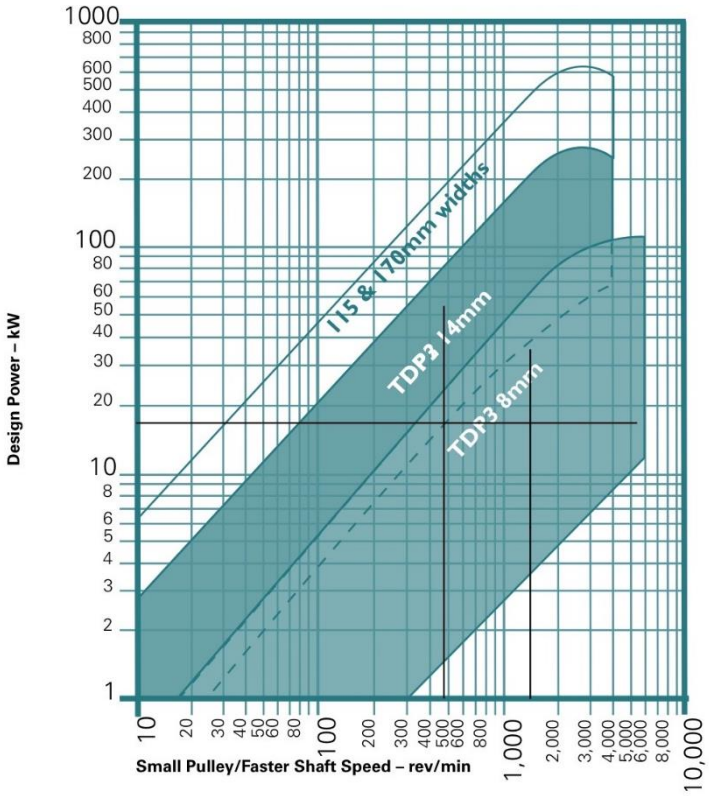


Figure 14: Graph from Fenner catalogue showing design power against the speed of the shaft.

Fenner has a wide range of options for a 3:1 reduction transmission system (Appendix F), but only two alternatives were considered due to their pulley centre distances. In the end, a centre distance of 533 was chosen as it was closer to the value specified in the design brief. This belt option would have delivered a power rating of 22.93 KW at a 20 mm belt width. This was sufficient for the design power required, but the driven pulley was only available at a belt width of 50. The belt was widened to accommodate for the pulley size leading to a power rating of 62.6 KW which is over 300% of the design power.

c. Bearings

When considering the bearings for the shaft design, the L_{10}life was calculated (630.7 millions of revolutions) (Appendix D), whilst referencing sub-assembly design specification No.21[1], so that the dynamic © and the static (c0) basic load ratings could be calculated at the two points where the bearings are located on the shaft (Table 9).

Table 9: Dynamic and static load ratings for Bearing A and B.

	Dynamic loading (kN)	Static Loading (kN)
Bearing A (ball)	19.52	2.28
Bearing B (roller)	20.50	2.39

Two single-row deep groove ball bearings for Bearings A and B were initially considered, with the inside diameters relating to the diameters of the shaft (40mm (SKF 6208) and 30mm (SKF 6206 ETN9) respectively). However, having two of these ball bearings would not allow for thermal expansion within the shaft, which could violate design demand No. 8[1]. Therefore, Bearing B was changed to a 30 mm single-row cylindrical roller bearing (NU 206 ECP). Additional calculations were carried out to ensure the roller bearings' suitability for the design demands, however, it's assumed that no radial forces act on the shaft (Appendix D). To conform with design demand No.20[1] a housing that could be bolted to the support structure for both bearings was created in Inventor. The roller bearings housing allowed for the associated thermal expansion of the shaft (Appendix G) by placing an internal circlip 1mm away from

the bearing inside the housing (Design Want No.23) [1]. A seal was also imported into the custom housing to prevent lubricants from dripping onto the floor and prevent contamination, to conform with Design Wants No.9 and 12[1]. To reduce the number of bespoke parts for this assembly an SKF Y-bearing unit could be used for the bearings. The use of the SKF Y-bearing units would also prevent bearing contamination and ensure lubricants do not drip due to seals and lubrication systems already in place.

d. Excel sheet and shaft dimensions

An Excel spreadsheet was formatted to test for failure at critical points. The two diagrams below show these critical points on the two design iterations being tested for failure.

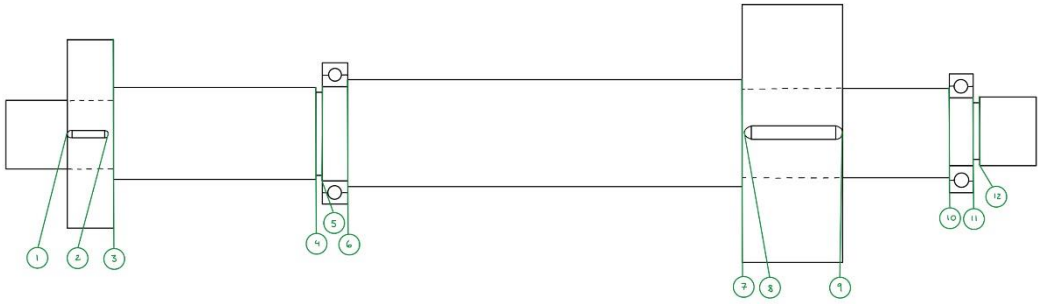


Figure 15: Critical points on shaft design 1

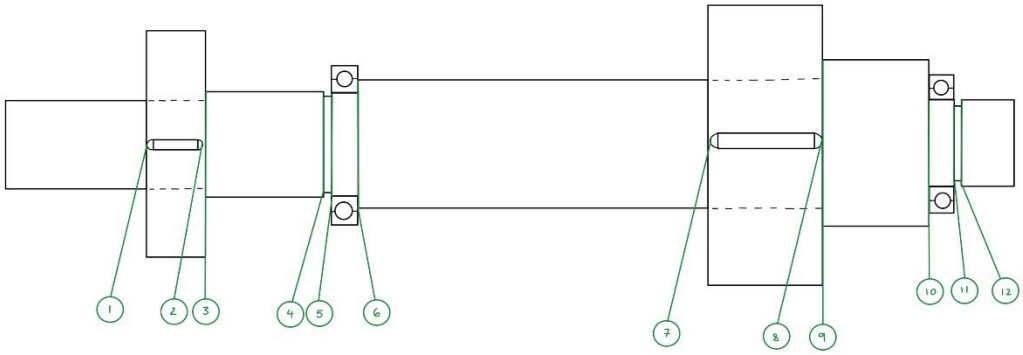


Figure 16: Critical points on shaft design 2

The first iteration of the diameters of the shaft in Figure (15) had to be increased at nodes 8, 9 and 10 as their allowed stress percentage was over 100%, this didn't influence any other components chosen as the taper locks and keys hadn't been calculated. The first iteration of node analysis of the shaft in Figure (16) was within the stress range, however, a problem was identified at node 10. The fillet radius had to be increased to 2mm to allow for the large decrease in shaft diameter, this would interfere with the construction of the shaft and impede the fit of the bearing on the shaft. Iteration 2 of the shaft in Figure (15) changed the values in a way to ensure the stress percentage was as close to 100% as possible, to reduce the weight and materials used for the shaft. The bearings were the first component that underwent design alterations however, due to the large steps in inside diameter, the next smallest bearings were not suitable for the design. The minimum bore size of the taper lock motor pulley bush also made a reduction of the shaft diameter at nodes 1 and 2 not viable.

e. Keys and keyways

To prevent relative rotation and enable torque transmission between the shaft and the taper locks a key and a keyway were implemented. Splines were also considered due to their capabilities of carrying large torques and their longer fatigue life, however, they take longer to manufacture making them the more expensive option. To conform with Design Want No15[1], minimising costs and manufacturing, calculations for keys were performed before spline calculations to avoid unnecessary details. Calculations using Von Misses equations to identify failure due to shear force and crushing from the

normal force were considered when selecting the minimum length of the key (Appendix E), with failure due to being crushed by the normal force being the larger of both. The minimum length at the motor pulley was 16.7mm and the minimum length of the key at the pump pulley was 8.51mm when using rectangular parallel keys (BS 4235-1.1972). The minimum lengths calculated due to failure were less than 2b therefore were increased to a length slightly smaller than the taper-locks they are constrained within. A normal fit is required as it allows for mass production assembly whilst not allowing the hub to slide over the key. The key should be manufactured from steel complying with BS 970, should not have a tensile stress of less than 550 MN/m^2 and align with the tolerances of rectangular parallel keys outlined in BS 4235-1:1972. Shingley's textbook gave initial estimates for the bending and torsional stress concentration factors for end-mill key seats of 1.7 and 3.0 which were used within the Excel file to ensure the location of the key seats didn't cause failure in the shaft. Multiple keys around the circumference were not needed to distribute the force carried by the keys based on the torque.

f. Circlips and taper locks

Circlips were used, in conjunction with shoulders, in the design to constrain the axial motion of the bearings along the shaft. The pulleys were constrained via a taper lock bush and shoulders to allow ease of assembly whilst axially locating the components on the shaft. DIN 471 D1 400 A40 from the Cirtex catalogue was used to constrain Bearing A. DIN 471 D1 400 A30 from the Cirtex catalogue was used to constrain Bearing B.

4. Conclusion

The final design iteration has been mathematically tested for failure and proven to withstand loading and stress conditions. Geometry, speed and operational requirements were abided by to create a shaft that reduced the RPM of a motor to a pump (1440 to 150 RPM). All design demands were met and wants were taken into consideration according to their priority to make the shaft suited to the customer's wishes. If this shaft design project were to be reproduced a higher preference should be placed on choosing from available steel grades, using standard SKF y-bearing units and reducing the length of sliding over tight fits. Applying calculations for lower yield strength steel grades could reduce the cost and embodied CO2 of the shaft, better conforming to Design Want No.7[1]. Using standard SKF Y-bearings would've been more time-efficient as seals and lubrication systems are already in place. Reducing the length of sliding over tight fits could've been reduced with different tolerances of the shaft before and after the circlip bearing, conforming better with Design Demands and Wants. More iterations and simulation calculations could be undertaken to produce a shaft that conforms perfectly to all design parameters.

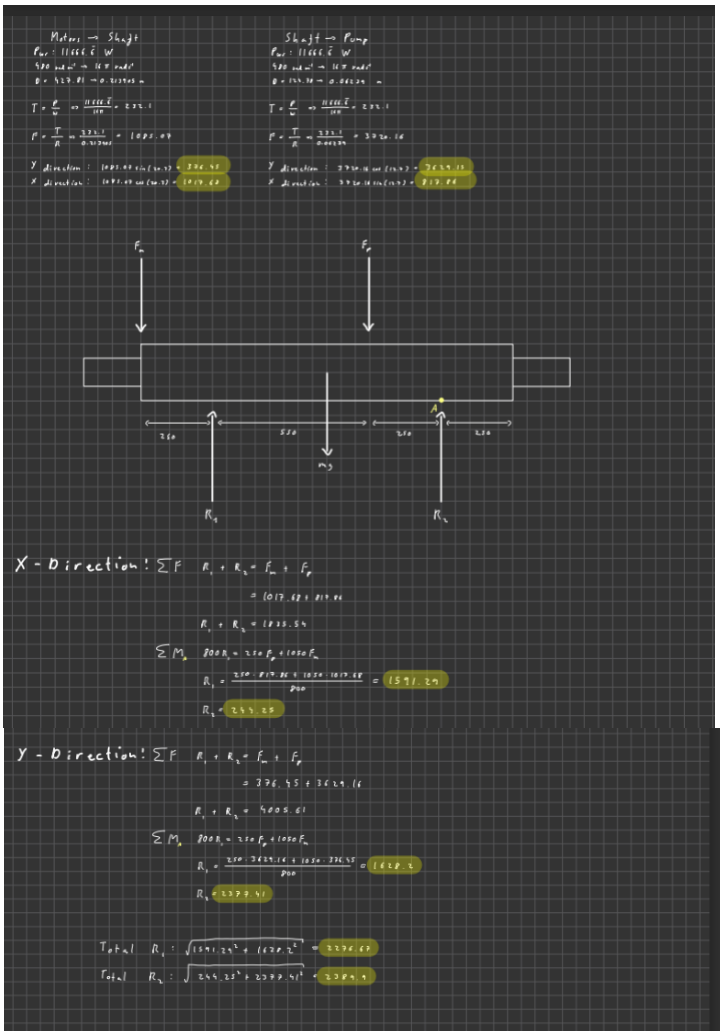
References

[1] Dr Lupton. Shaft Design 2023 – pump layshaft design brief[Design brief]. ME20018 Design 3 University of Bath; [Cited Nov 7 2023]. Available from: https://bath-prod-sss1.s3.eu-west-1.amazonaws.com/69/e8/69e8ff434a1dc3b86a52b2adb6dcbdcbec273379?response-content-disposition=inline%3B%20filename%3D%22Shaft%20Design%202023%20-%20pump%20layshaft%20design%20brief.pdf%22&response-content-type=application%2Fpdf&X-Amz-Content-Sha256=UNSIGNED-PAYLOAD&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAJBFBMNJTZPM2NVZA%2F20231109%2Feu-west-1%2Fs3%2Faws4_request&X-Amz-Date=20231109T104737Z&X-Amz-SignedHeaders=host&X-Amz-Expires=21563&X-Amz-Signature=074c8328b385e694d98627b08a0770eca514818da1886955686d694daf74a4d9

[2] Dr Gopsill ,Dr Lupton, Dr Owen, Prof. Mullineux. D E TA I L E D S U B - A S S E M B LY D E S I G N. FA C U LT Y O F E N G I N E E R I N G A N D D E S I G N | U N I V E R S I T Y O F B AT H , U K [cited Nov 7 2023]. Available from: https://bath-prod-sss1.s3.eu-west-1.amazonaws.com/b2/b0/b2b0148e270fd700ad4922f010654c5eb6fdb682?response-content-disposition=inline%3B%20filename%3D%22Design3-subassembly-course-notes.pdf%22&response-content-type=application%2Fpdf&X-Amz-Content-Sha256=UNSIGNED-PAYLOAD&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAJBFBMNJTZPM2NVZA%2F20231109%2Feu-west-1%2Fs3%2Faws4_request&X-Amz-Date=20231109T104825Z&X-Amz-SignedHeaders=host&X-Amz-Expires=21575&X-Amz-Signature=a45b65ebcc562c11517da46594fd1571c3fbbc47d23767fd9e5afa84932b075f

Appendix

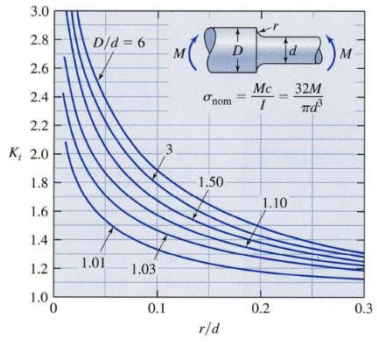
Appendix A: calculations of shaft loading conditions



Appendix B: Excel sheet containing failure test equations for final design iteration.

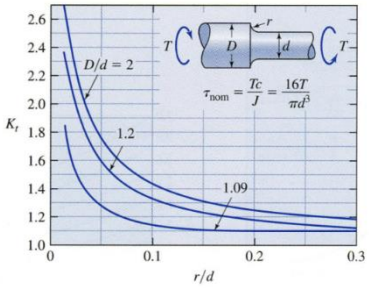
Node				1	2	3	4	5	6	7	8	9	10	11	12
Description of point			Example	Motor Key Beginning	Motor Key End	Sholder 1	Bearing 1 Circlip Beginning	Bearing 1 Circlip End	Sholder 2	Sholder 3	Pump Key Beginning	Pump Key End	Sholder 4	Bearing 2 Circlip Beginning	Bearing 2 Circlip End
Distance along shaft				(-0.03)	0.03	0.03	0.23925	0.241	0.259	0.835	0.835	0.87	1.042	1.058	1.0595
Bigger Diameter				0.035	0.035	0.04	0.04	0.04	0.045	0.045	0.04	0.04	0.037	0.03	0.03
Smaller Diameter	d	m	0.028	0.035	0.035	0.035	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.03	0.03
Vert BM	Mv	Nm	0	0	11.29	11.29	90.07	90.72	82.85	511.14	511.14	427.93	19.01	0	0
Horiz BM	Mh	Nm	0	0	30.53	30.53	243.48	245.26	249.26	52.52	52.52	43.97	1.96	0	0
Combined BM	M	Nm	142.4	0	32.55	32.55	259.61	261.50	262.67	513.83	513.83	430.18	19.11	0	0
Torque	Q	Nm	124.3	232.1	232.1	232.1	232.1	232.1	232.1	232.1	232.1	232.1	232.1	232.1	232.1
Bending SCF	Km		1.68	2.1	2.1	2.19	5	5	2.22	2.22	2.1	2.1	2.26	5	5
Torsion SCF	Ks		1.42	3	3	1.71	3	3	1.74	1.74	3	3	1.75	3	3
Bending stress, nominal	σ_n	Pa	66074821.86	0	7733136	7733136	41317529	41619120	41804969	81778767	81778767	68465757	7209657	0	0
Torsion stress, nominal	τ_n	Pa	28838133.27	27570276.23	27570276	27570276	18469931	18469931	18469931	18469931	18469931	18469931	43780578	43780578	43780578
Combined stress, nominal	σ'_n	MPa	82829921.38	47.75	48.38	48.38	52.25	52.49	52.64	87.81	87.81	75.57	76.17	75.83	75.83
Bending stress	σ	MPa	111005700.7	0	16.24	16.94	206.59	208.10	92.81	181.55	171.74	143.78	16.29	0.00	0.00
Torsion stress	τ	MPa	40950149.25	82.71	82.71	47.15	55.41	55.41	32.14	32.14	55.41	55.41	76.62	131.34	131.34
Combined stress	σ'	Mpa	131730823.1	143.26	144.18	83.40	227.79	229.16	108.22	189.89	196.73	172.87	133.70	227.49	227.49
Fail or No Fail			131730553.9	53.21	53.55	30.98	84.61	85.12	40.20	70.53	73.07	64.21	49.66	84.50	84.50

Appendix C: SKF's for shoulders



Approximate formula
$K_t \approx B\left(\frac{r}{d}\right)^a$, where:

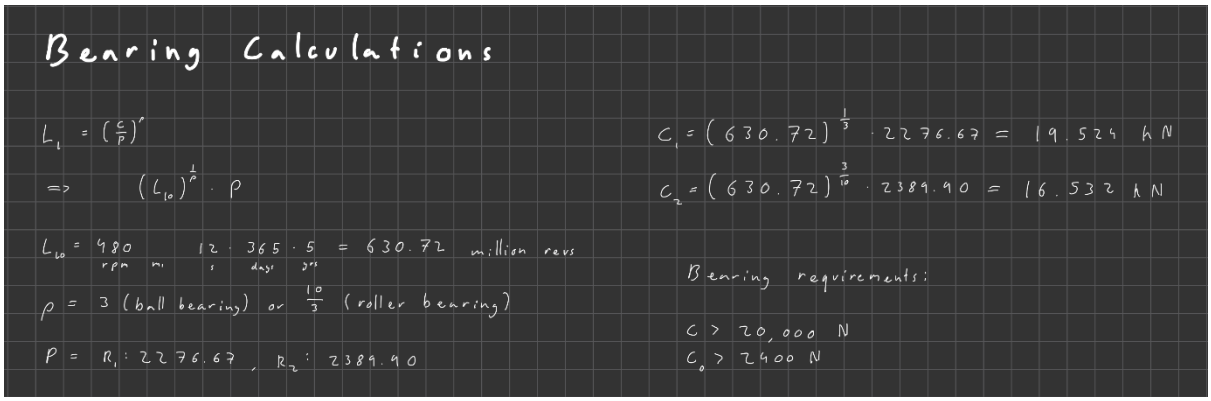
D/d	B	a
6.00	0.879	−0.332
3.00	0.893	−0.309
1.50	0.938	−0.258
1.10	0.951	−0.238
1.03	0.981	−0.184
1.01	0.919	−0.170



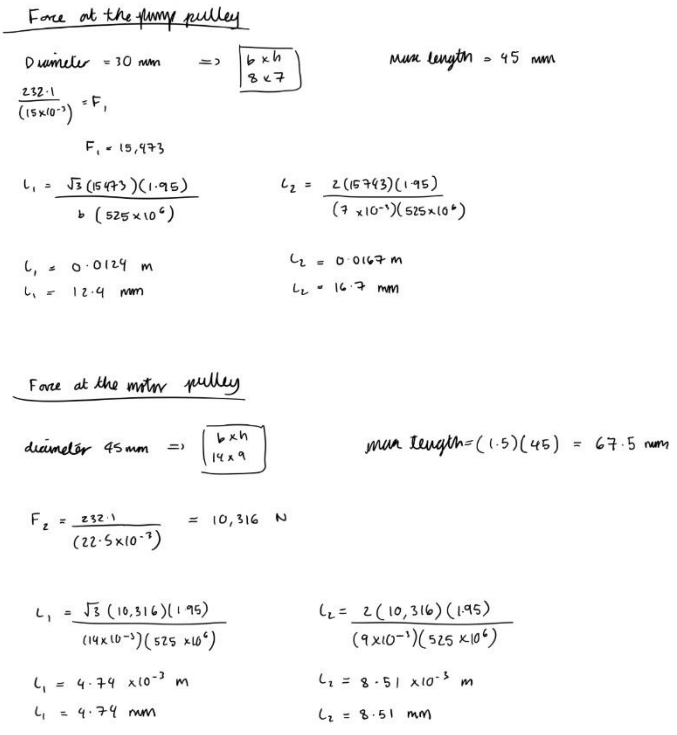
Approximate formula
$K_t \approx B\left(\frac{r}{d}\right)^a$, where:

D/d	B	a
2.00	0.863	−0.239
1.20	0.833	−0.216
1.09	0.903	−0.127

Appendix D: Bearing calculations



Appendix E: calculations for the minimum key lengths



Appendix F: Fenner Torque Drive Plus synchronous belts table.

Fenner Torque Drive Plus® *3* 8MXP & HTD 8M Drives

CENTRE DISTANCE IN MILLIMETRES

Speed Ratio	Number of grooves on Driving Pulley	Number of grooves on Driven Pulley	Belt pitch length in millimetres 480 (60 teeth)	560 (70 teeth)	600 (75 teeth)	640 (80 teeth)	720 (90 teeth)	800 (100 teeth)	880 (110 teeth)	960 (120 teeth)	1040 (130 teeth)	1120 (140 teeth)	1200 (150 teeth)	1280 (160 teeth)	1440 (180 teeth)	1600 (200 teeth)	1760 (220 teeth)	1800 (225 teeth)	2000 (250 teeth)	2400 (300 teeth)	2600 (325 teeth)	2800 (350 teeth)	Speed Ratio
2.50	32	80	–	–	–	–	–	165	207	248	290	330	371	411	492	573	653	673	774	974	1074	1174	2.50
2.50	36	90	–	–	–	–	–	–	–	217	259	300	341	382	463	544	624	644	745	946	1046	1146	2.50
2.55	44	112	–	–	–	–	–	–	–	–	–	232	274	316	399	480	561	582	683	884	984	1085	2.55
2.57	28	72	–	–	–	–	150	192	233	274	315	356	396	436	517	597	678	698	798	998	1099	1199	2.57
2.57	56	144	–	–	–	–	–	–	–	–	–	–	–	–	299	384	467	487	589	792	893	994	2.57
2.63	64	168	–	–	–	–	–	–	–	–	–	–	–	–	–	–	394	415	519	724	825	927	2.63
2.65	34	90	–	–	–	–	–	–	178	220	262	304	345	385	467	547	628	648	749	949	1050	1150	2.65
2.67	24	64	–	–	–	134	177	218	259	300	340	381	421	461	542	622	702	722	822	1023	1123	1223	2.65
2.67	30	80	–	–	–	–	–	168	210	252	293	334	375	415	496	576	657	677	777	978	1078	1178	2.67
2.67	72	192	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	446	654	757	858	2.67
2.77	26	72	–	–	–	–	153	195	237	278	319	359	400	440	521	601	681	702	802	1002	1102	1203	2.77
2.80	40	112	–	–	–	–	–	–	–	–	–	238	281	323	406	487	569	589	690	891	992	1092	2.80
2.81	32	90	–	–	–	–	–	–	181	224	266	307	348	389	470	551	632	652	752	953	1053	1154	2.81
2.86	28	80	–	–	–	–	–	171	214	255	297	338	378	419	500	580	661	681	781	982	1082	1182	2.86
2.95	38	112	–	–	–	–	–	–	–	–	–	242	284	326	409	491	572	593	694	895	996	1096	2.95
3.00	24	72	–	–	–	–	156	199	240	281	322	363	403	444	524	605	685	705	806	1006	1106	1206	3.00
3.00	30	90	–	–	–	–	–	–	184	227	269	311	352	393	474	555	635	656	756	957	1057	1157	3.00
3.00	48	144	–	–	–	–	–	–	–	–	–	–	–	–	312	397	480	501	604	807	908	1009	3.00
3.00	56	168	–	–	–	–	–	–	–	–	–	–	–	–	–	320	407	428	533	738	840	941	3.00
3.00	64	192	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	350	459	668	771	873	3.00
3.08	26	80	–	–	–	–	–	174	217	259	300	341	382	422	503	584	664	685	785	986	1086	1186	3.08

Appendix G: Thermal expansion calculations

$$\Delta \ell = L \alpha \Delta T$$

$$\Delta \ell = (65)(11 \times 10^{-6})(1106 \times 10^{-3})$$

$$\Delta \ell = 0.79 \text{ mm}$$